

BERT vs TF-IDF: Sentiment Classification on SST-2

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1 What is this project about

In this project I built two models that can tell if a sentence has a positive feeling or a negative feeling. This is called sentiment classification. I used a dataset called SST-2, which is part of a well known benchmark called GLUE. Each sentence in this dataset has a label: negative or positive.

I built two models:

- A simple model using TF-IDF and Logistic Regression. This is a classic method that has been used for a long time.
- A model using BERT, which is a deep learning model that understands language better because it looks at words in their full context.

The goal was to compare these two models and see how much better BERT is, and also to understand why.

2 The dataset

SST-2 has short sentences taken from movie reviews. Each sentence is labeled as positive or negative.

- Training set: 67,349 sentences
- Test set: 872 sentences (428 negative, 444 positive)

The training set is quite big, so both models had enough data to learn from.

3 The simple model: TF-IDF and Logistic Regression

The first model uses TF-IDF. This is a way to turn text into numbers based on how important each word or phrase is. I used unigrams, bigrams, and trigrams, which means single words, pairs of words, and groups of three words. I kept the top 50,000 most useful features.

Then I used Logistic Regression, which is a simple and fast machine learning model, to classify the sentences based on these numbers.

The results on the test set were:

- Accuracy: 0.8108 (about 81 percent)
- Macro F1 score: 0.8101

This means the model got about 81 out of 100 sentences correct. This is a good baseline, but there is room for improvement.

4 The deep learning model: BERT

The second model uses BERT, specifically the version called bert-base-uncased. BERT already knows a lot about language because it was trained on a huge amount of text before I used it. I just needed to fine tune it for this specific task.

I trained it for 4 epochs, which means the model went through the whole training set 4 times. Since SST-2 sentences are short, I set the maximum length to 64 tokens, which was enough for almost all sentences.

Training took about 17 minutes using a GPU on Google Colab.

The results on the test set were:

- Accuracy: 0.9266 (about 93 percent)
- Macro F1 score: 0.9265

This is a big improvement over the simple model.

5 Comparing the two models

Table 1: Accuracy and Macro F1 for both models

Model	Accuracy	Macro F1
TF-IDF + Logistic Regression	0.8108	0.8101
BERT fine tuned	0.9266	0.9265
Difference (BERT minus baseline)	+0.1158	+0.1165

BERT is about 11.6 points better than the simple model, both in accuracy and in Macro F1. This is a clear and important improvement.

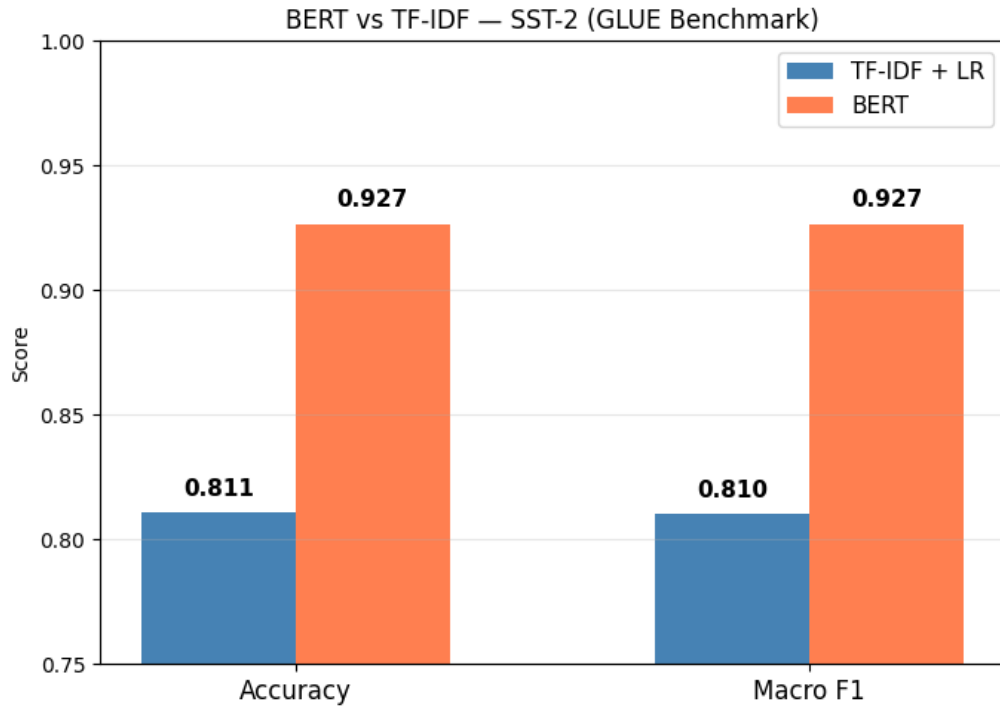


Figure 1: This chart shows the accuracy and Macro F1 score for both models side by side. The orange bars (BERT) are much higher than the blue bars (TF-IDF), showing that BERT performs better on both measures.

6 Looking closer with confusion matrices

A confusion matrix shows exactly how many sentences were classified correctly and how many were mistaken for the other class. This helps us see if a model has trouble with one class more than the other.

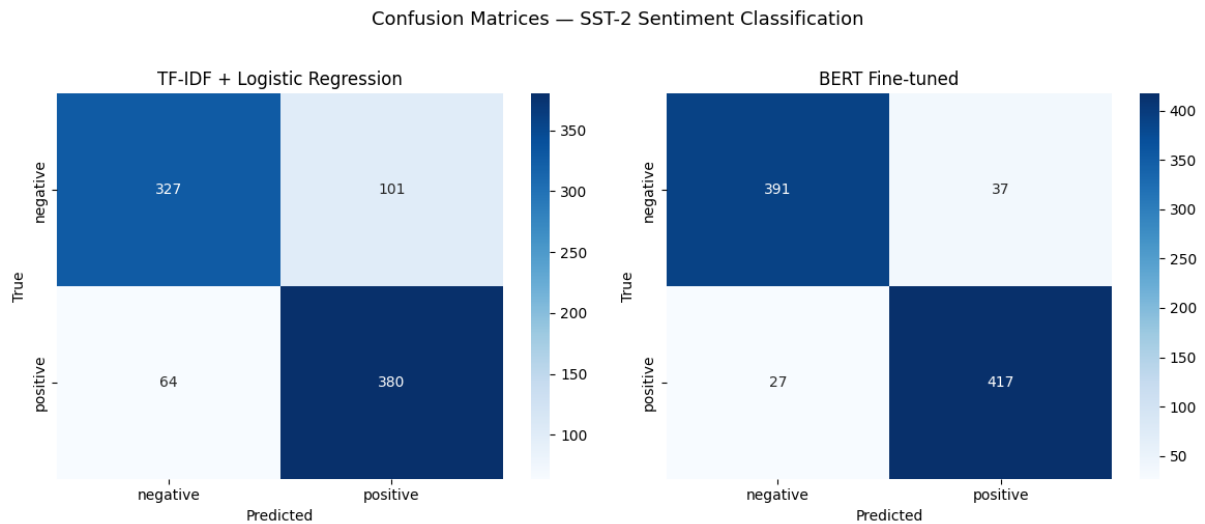


Figure 2: On the left, the TF-IDF model confused 101 negative sentences as positive, and 64 positive sentences as negative. On the right, BERT made far fewer mistakes: only 37 negative sentences were called positive, and only 27 positive sentences were called negative. BERT is more balanced and makes fewer errors in both directions.

If we look at the numbers:

- TF-IDF model: out of 428 negative sentences, it got 327 correct. Out of 444 positive sentences, it got 380 correct.
- BERT model: out of 428 negative sentences, it got 391 correct. Out of 444 positive sentences, it got 417 correct.

BERT is better at recognizing both negative and positive sentences. This means it understood the sentences in a deeper way, not just by spotting certain keywords.

7 Looking at the training process

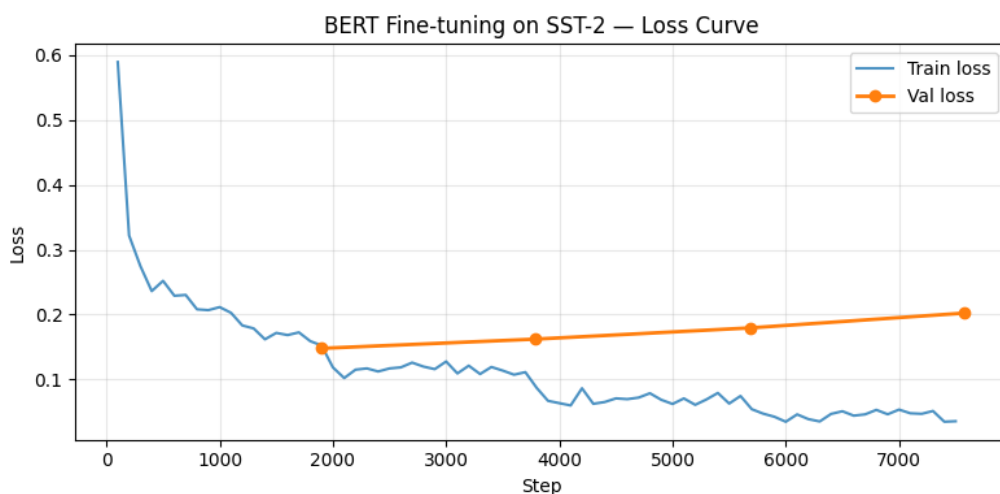


Figure 3: This chart shows the loss during training. The blue line is the training loss, and it goes down quickly at first and keeps getting smaller. The orange line is the validation loss, and it goes down a little at first, but then slowly goes up again after around step 2000. This means the model still gets better at the main task, but it starts to memorize the training data a bit too much in later steps.

This pattern is called overfitting. It happens when a model learns the training data so well that it starts to lose a little bit of its ability to handle new, unseen data. However, even with this small overfitting, the final accuracy and F1 score on the test set were still very high, so the model still performed very well overall.

8 Why does BERT do better

There are a few reasons why BERT works better than the simple TF-IDF model on this task.

- **Context matters.** TF-IDF looks at words and short phrases without really understanding the full meaning of a sentence. BERT looks at the whole sentence at once and understands how words relate to each other.
- **Negation is hard for simple models.** A sentence like "not as bad as expected" has the word "bad" in it, which might confuse a TF-IDF model into thinking it is negative. BERT can understand that "not as bad" actually means something more positive.
- **BERT already knows a lot.** Before I even trained it, BERT had already learned a lot about how language works from a huge amount of text. I only needed to teach it the

specific task of sentiment classification, which is faster and gives better results.

9 Limitations and possible improvements

This project gave good results, but there are some things that could be improved in the future:

- I only used one random seed. Running the experiment a few more times with different seeds would show if the results are stable.
- I did not search for the best settings (hyperparameters) for either model. Trying different values might improve both models a little.
- This task only has two classes, positive and negative. It would be interesting to try a version of this dataset with more classes, like very positive, slightly positive, neutral, slightly negative, and very negative.

10 Conclusion

In this project, I compared a classic machine learning approach (TF-IDF with Logistic Regression) with a modern deep learning approach (fine tuned BERT) for sentiment classification on the SST-2 dataset.

The simple model reached about 81 percent accuracy, while BERT reached about 93 percent accuracy, an improvement of more than 11 points. BERT also made fewer mistakes for both negative and positive sentences, showing that it understands sentences in a deeper and more balanced way.

This project helped me understand not just how to use these models, but also why deep learning models like BERT can perform so much better than older methods on language tasks.